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Supplemental information

Human anti-glycan reactivity emerges from B cells utilizing private gene rearrangements that are affinity maturated in germinal centers

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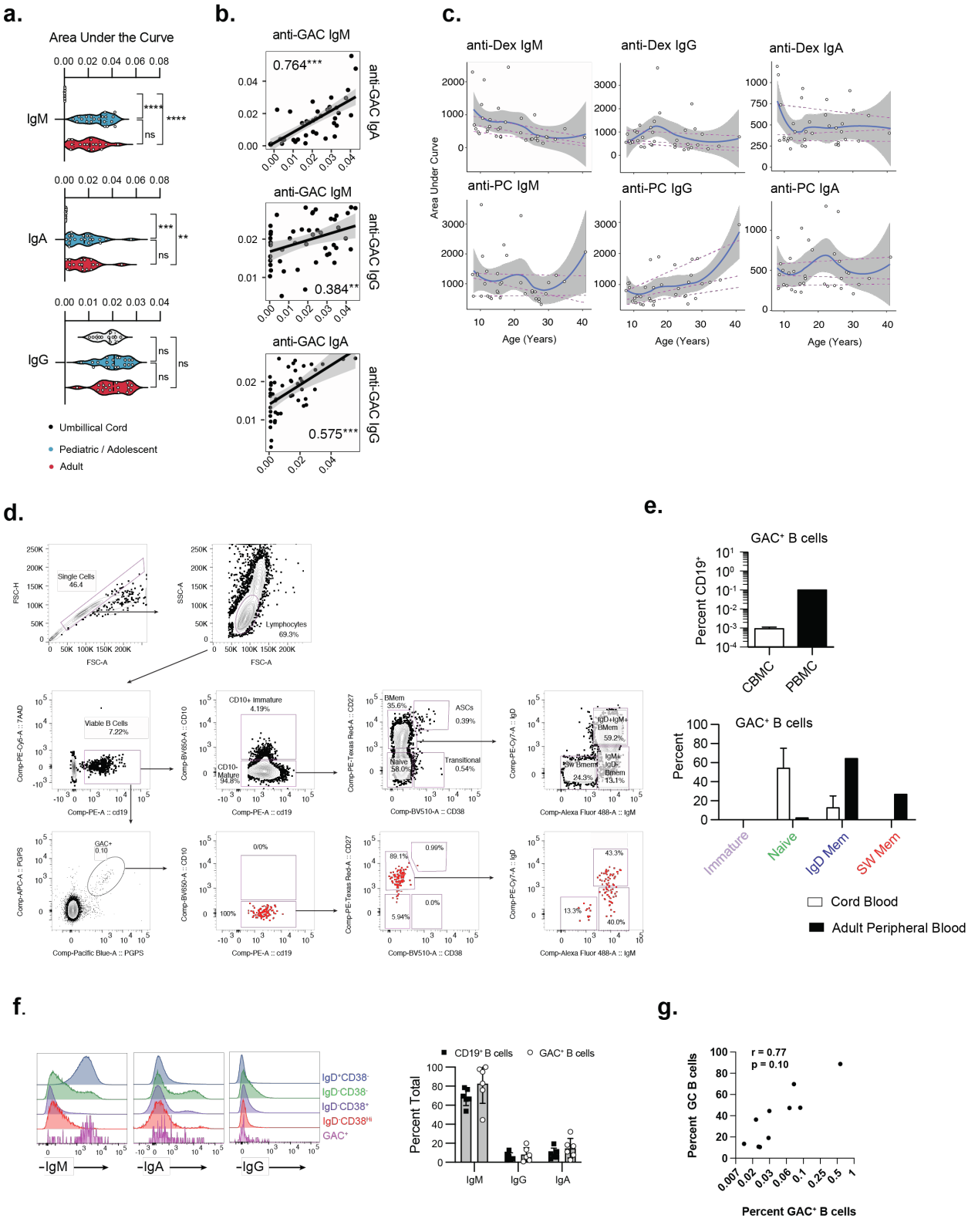


Figure S1. Emergence of nAb reactivities and GAC-reactive B cells, Related to Figure 1.
(a) Heat map of the area under the curve absorbance values of serial dilutions of sera from the indicated age groups binding GAC measured by ELISA.

(b) Correlation area under the curve values of GAC-reactive Ab of the indicated isotypes.
(c) Area under the curve values of serum Ab of the indicated isotype binding dextran (polymers of α1-3 glucose, upper) or phosphorylcholine (PC, lower) plotted by donor age.
(d) Gating scheme of flow cytometric histograms of adult blood to define B cell subpopulations enumerated in Figure 1d.
(e) Frequency of total GAC^+ B cells (upper) and subset frequencies of GAC^+ B cells (lower) from the indicated specimens.
(f) Representative histograms of isotype expression of the indicated phenotype of tonsillar B cells (left) and enumeration of GAC-binding B cells expressing the indicated isotype (right).
(g) Correlation of the frequency of GAC-binding B cells (plotted as log2) and the frequency of germinal center B cells.

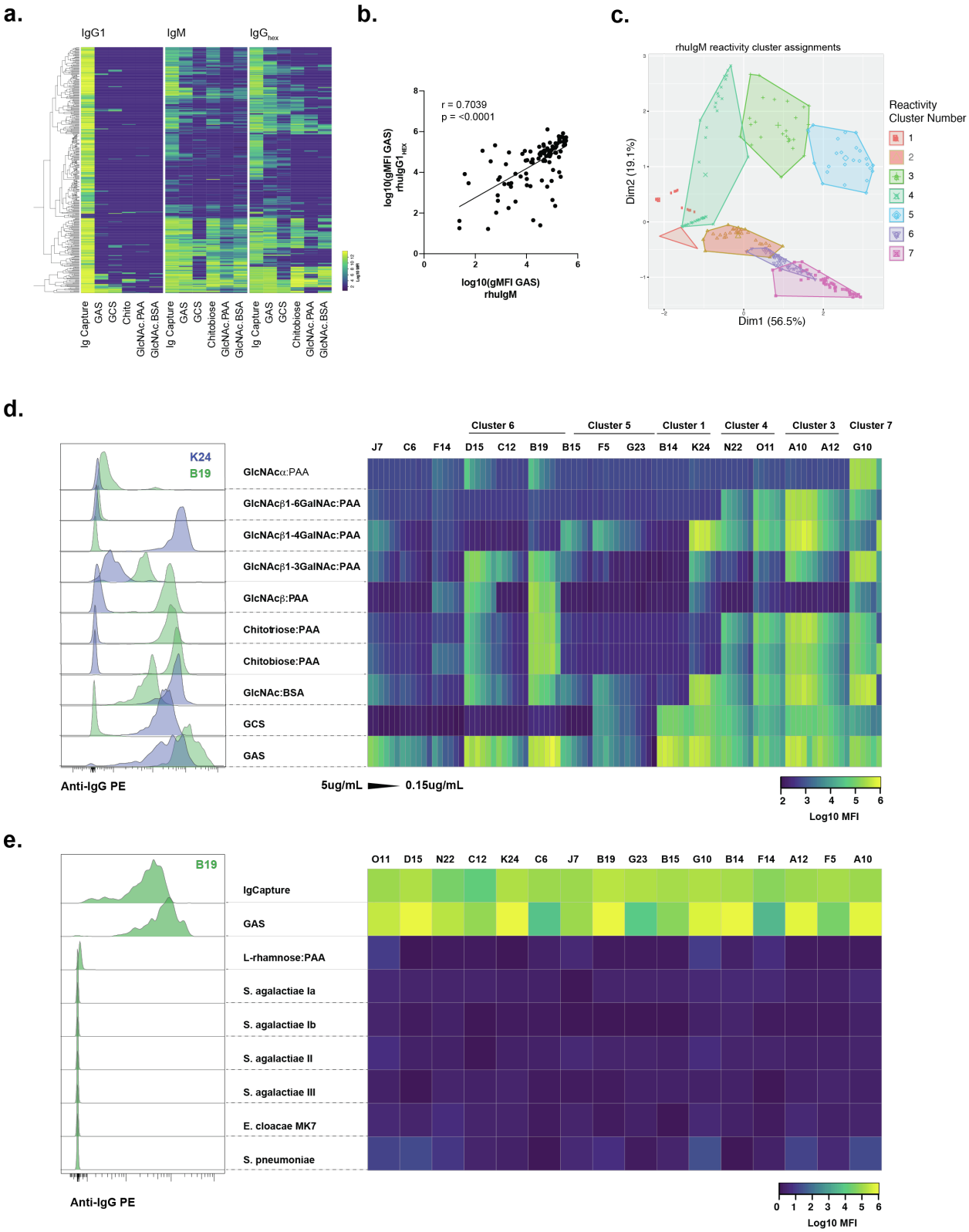


Figure S2. GAC-reactive Abs benefit from polyvalency and display discrete reactivity profiles against GlcNAc antigens, Related to Figure 2.
(a) PCoA plot of Kmeans clustering of MFI tables of rhIgM in Figure 2.

(b) Heatmap of GlcNAc array gMFI tables of a subset of tonsil GAC^{+} B cell derived rAbs expressed as the indicated constructs.
(c) Correlation of GAC-binding rAbs expressed as IgM or $IgG1_{HEX}$.
(d) Cytometric histograms of the listed antigens stained with the indicated rmAbs (0.4ug/mL) (left) and heat map of the MFIs of the antigens resulting from serial dilutions of the indicated mAbs.
(d) Cytometric histograms of the indicated bacteria or L-rhamnsose and Ig capture beads stained with the rmAb B19 (0.4ug/mL) (left) and heat map of the MFIs of the antigens resulting from serial dilutions of the indicated mAbs.

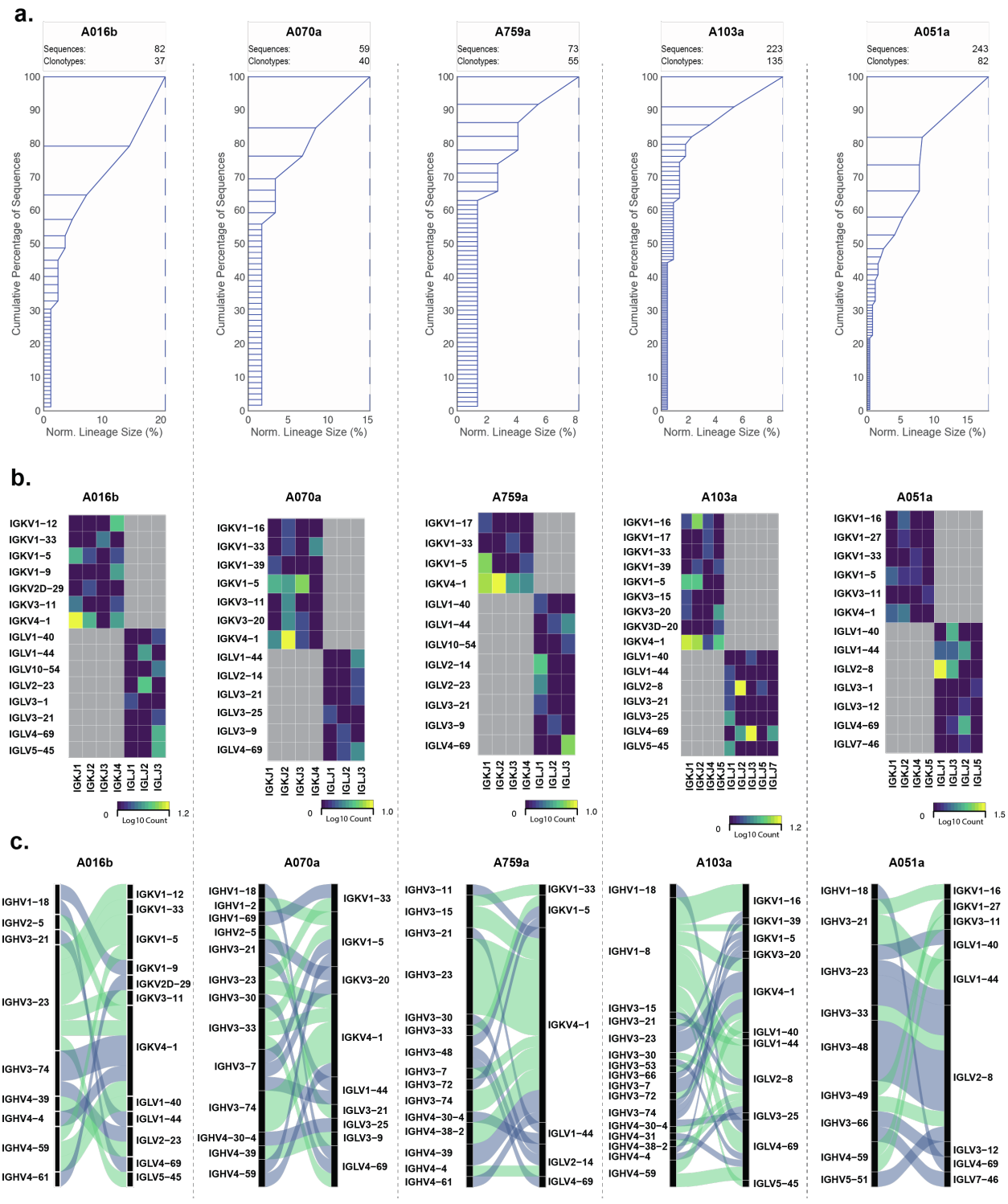


Figure S3. GAC$^+$ B cell clonotypes are expanded in human tonsil, Related to Figure 3.
(a) Clonality plots of BCR receptor sequences derived from GAC$^+$ B cells sorted from the indicated specimens. Number of sequences recovered, lineage sizes, and cumulative percentages of each clonotype are shown.
(b) Heatmap of the number of recovered light chains utilizing the indicated IGKV or IGLV paired with their associated IGKJ and IGLJ genes.

(c) Alluvial plots of IGHV and IGK/LV pairs within mAbs cloned from the indicated tonsil GAC^{+} B cells.

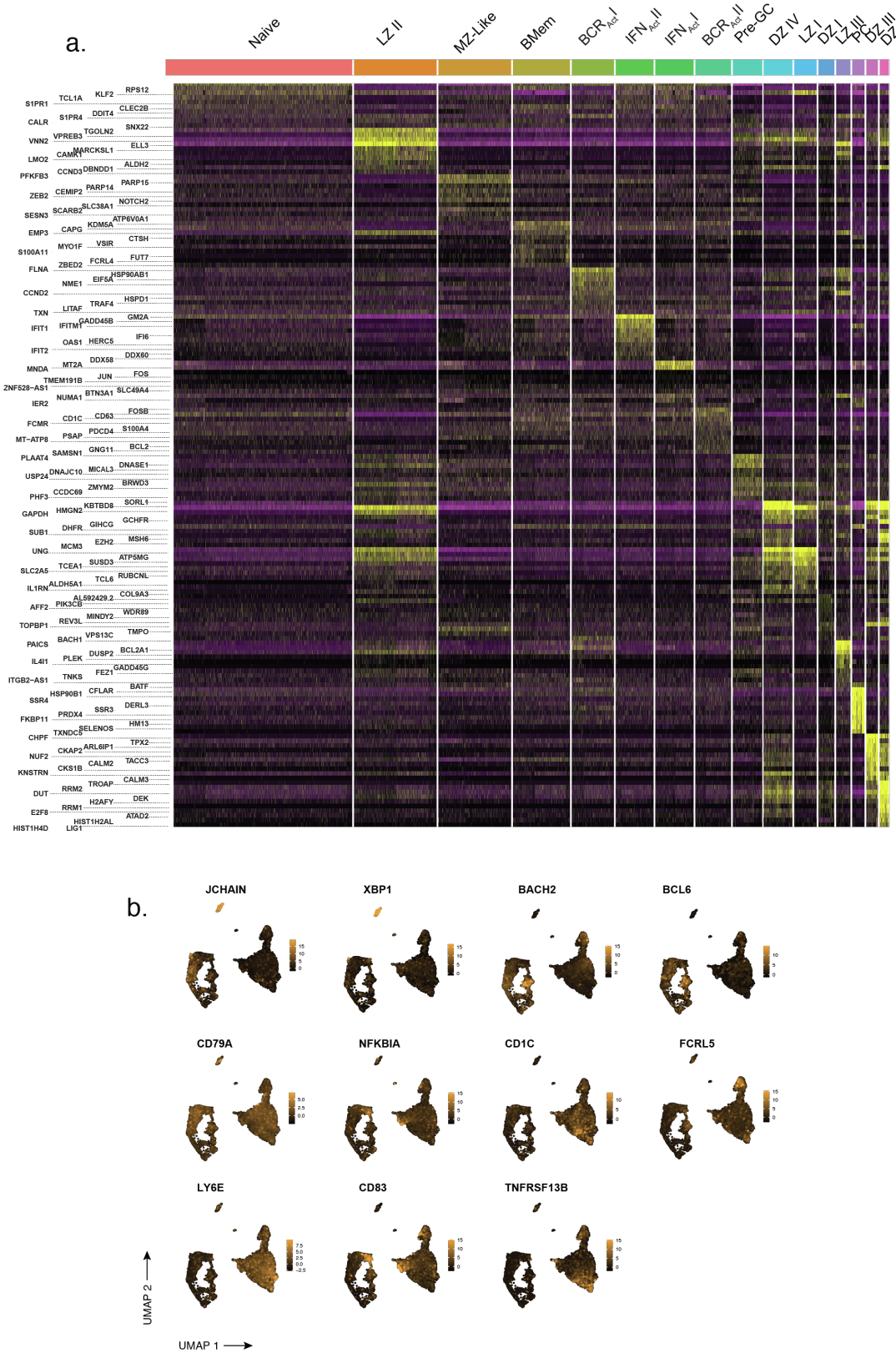


Figure S4. GAC^{+} B cells are distributed through multiple B cell subsets but enriched in GC, Related to Figure 4.

(a) Heat map of the expression of differentially expressed genes used to annotate cluster identity.
(b) Feature plots of indicated genes in UMAP space.

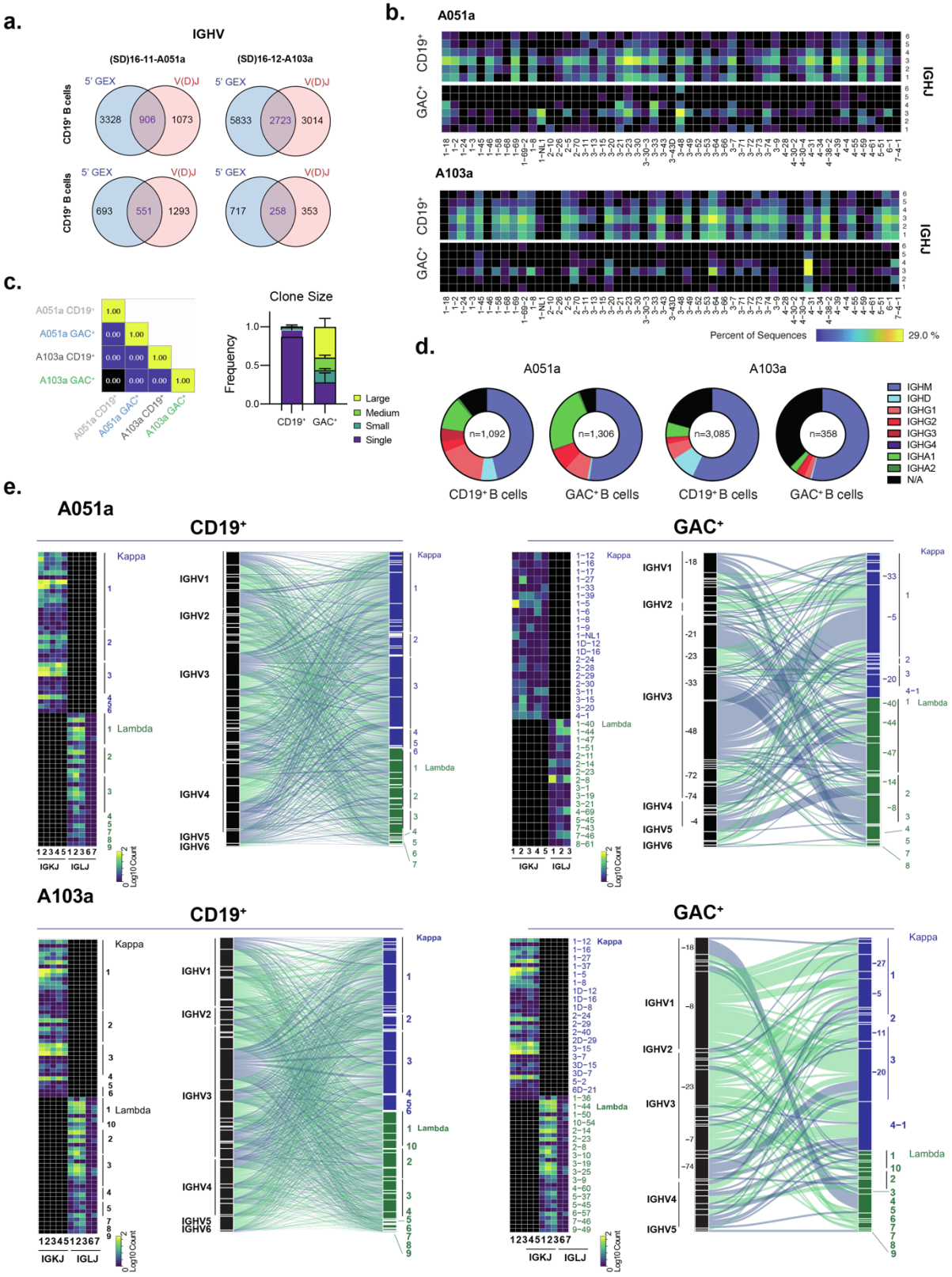


Figure S5. GAC⁺ B cells are expanded and diversified in tonsil, Related to Figure 5.

(a) Venn diagrams of the filtered 10x data base. The number of cell barcodes isolated in 5' GEX and VDJ sequencing libraries and overlapping cells are shown for each sample.
(b) Heat map of IGHV and IGHJ gene usage of GAC^{+} B cell receptors of the 10x VDJ sequences derived from the indicated tonsil specimens.
(c) Heat map of the Morissita-Horn overlap index for each population of both 10x VDJ libraries
(d) Donut plots of the Ig isotype distribution within IGHV sequences from the indicated sorted populations. Isotypes are depicted in by the indicated colors.
(e) Heatmap of light chain V and J gene usage (left) and alluvial plots of IGHV and IGK/LV pairing (right) derived from the 10x VDJ sequences from the indicated B cell populations.

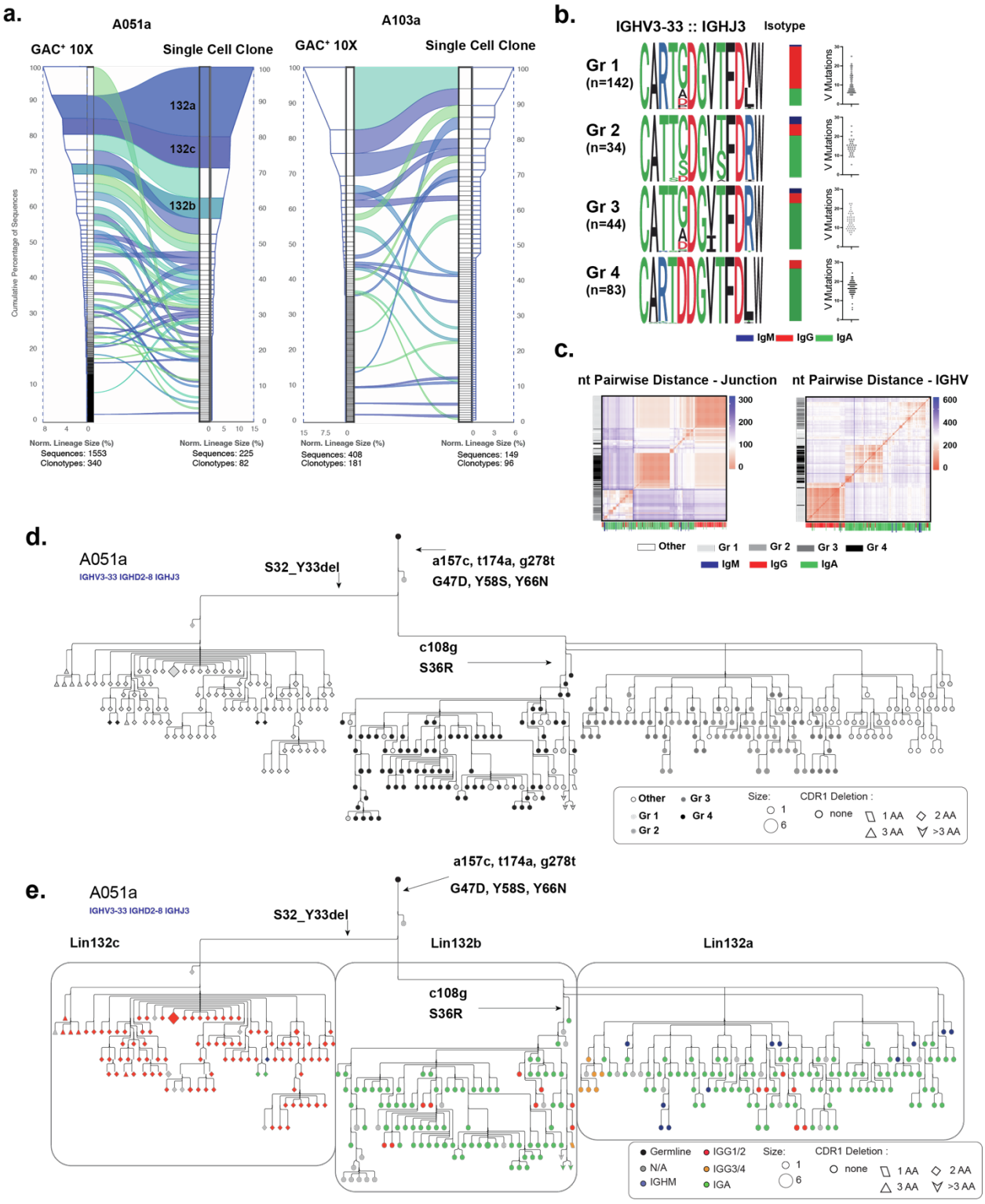


Figure S6. Concordance between single-cell and 10x VDJ sequence recovery, Related to Figure 6.

(a) Clonality alluvial diagrams of lineages recovered from GAC$^+$ cells in 10x VDJ libraries (left) and single cell sorting and sequencing (right). Cumulative percentages on outer clonality plots indicate percentage of total sequences of each clone. Ribbons connect shared lineages.

Lineages related to Lin132 are highlighted. The number of sequences and unique lineage clonotypes are shown
(b). Amino acid sequence logos of IGH VDJ junctions of four clonotypes identified in A051a derived IGHV3-33:IGHJ3 clonotypes Isotype distribution (bar plots) and somatic mutation rates (scatter plots) of members of the indicated groups are shown.
(c) Distance matrix heat maps of the pairwise junction nt (left) or IGH region (right). Lineage group assignments are plotted on the x axis and isotypes on the y axis.
(d) Phylogenic tree of nt parsimony of all members of groups 1-4. Somatic mutations corresponding to early branch points are indicated. Each sequence is colored in grey scale corresponding to its group membership. Shapes indicate number of AA deletions in CDR1 of each sequence.
(e) Phylogenic tree of nt parsimony of all members of groups 1-4. Sequences are colored by originating isotype.

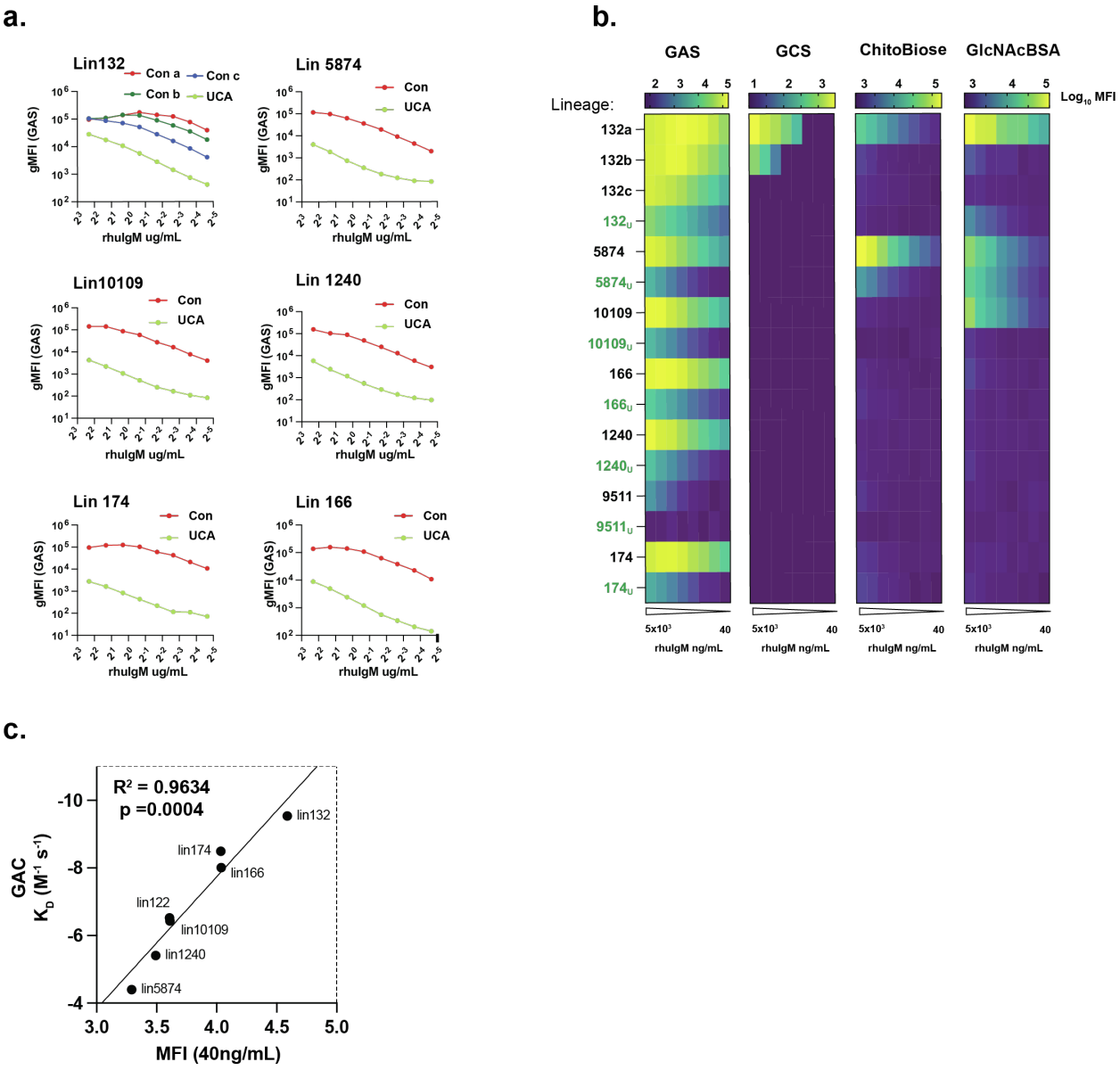


Figure S7. Expanded GAC$^+$ B cells expanded B cell lineages are derived from GAC$^+$-reactive precursors and display heterogenous binding to GlcNAc, Related to Figure 7.
(a) Dilution curves of the MFI resulting from staining GAS with the rhIGM derived from the indicated lineage consensus (red) or UCA (green) at the indicated dilutions.
(b) Heatmap of the MFI resulting from staining of the indicated bioparticle/antigen coated beads at the indicated concentration. rhIgM derived from lineage consensus sequence are labeled in black, UCAs denoted by subscript U in green.
(c) Correlation of the SPR generated K_{Dapp} of the indicated rhIgM and the MFI of GAS stained at 40ng/mL.